

Phoebe Kim

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EDUCATION

The University of Texas at Austin

Bachelor of Science, Computer Science

Austin, TX

Expected May 2028

- GPA: 3.7/4.0
- Relevant Coursework: Data Structures, Algorithms, Operating Systems, Computer Architecture, Computer Graphics, Ethical AI, Probability, Statistics, Cloud Native Computing, Agentic Software Engineering

EXPERIENCE

KLA

Austin, TX

Software Engineer Intern (OVALiS Team, Part-time)

Aug 2026 - Present

- Fixed wafer plot recalculation issues when switching between overlay controller strategies, ensuring plots correctly refresh after strategy selection changes
- Implemented automated tests for overlay metrics bias computation, validating new lot-level calculations grouped by context parameters and updating existing suites for UI changes

Software Development Engineer in Test Intern (OVALiS Team)

May 2026 - Aug 2026

- Built and deployed an MCP server integrating Aqua Cloud REST APIs for use across QA and development teams
- Developed a Windsurf AI workflow that leveraged MCP tools to generate and upload test cases to Aqua
- Automated unit tests for newly developed features in OVALiS using existing test automation framework, expanding regression test coverage

Nuclear Engineering Teaching and Research Lab

Austin, TX

Research Assistant at The University of Texas at Austin

Feb 2025 - Feb 2026

- Designed PostgreSQL schemas storing high-volume reactor simulation outputs and metadata across thousands of runs
- Built a text to SQL chatbot using LangChain to generate PostgreSQL queries from natural language questions and implemented interactive Plotly visualizations enabling users to graph query results directly
- Deployed lab website on cloud VMs (EC2) and validated performance with external users

PROJECTS

PintOS — Operating Systems Project

2025

- Implemented kernel multithreading, synchronization primitives, and priority scheduling to resolve race conditions and priority inversion
- Built a virtual memory subsystem supporting demand paging and page-fault handling across concurrent kernel workloads; designed a filesystem buffer cache improving concurrent disk I/O efficiency

Aqua MCP Server for QA Automation — KLA Intern Project

2026

- Developed 20+ MCP tools to automate QA workflows (research previous tests, write new tests, analyze test coverage), reducing manual test setup time from one week to 1-2 days for the QA team
- Implemented asynchronous execution and multithreaded background database synchronization, enabling non-blocking tool execution and improving responsiveness for concurrent QA operations
- Implemented a semantic search system using ChromaDB and vector embeddings to retrieve similar test cases

Personalized AI Assistant (“My Virtual Puppet”) — [Full-Stack AI Application](#)

2025

- Fine-tuned LLaMA on personal data using 8-bit quantization, achieving 2x faster inference with 70% lower VRAM usage
- Deployed a full-stack AI chatbot with React and Three.js, featuring a 3D interface and a secure Netlify backend

SKILLS

Languages: Python, Java, C/C++/C#, TypeScript, SQL

Frameworks & AI: React, Flask/FastAPI, Node.js, PyTorch, LangChain, REST APIs

Infrastructure & Platforms: PostgreSQL, Docker, Kubernetes, AWS, Azure, MCP, Vector Search, Linux, Git